LEZQ

COURSE

INTRODUCTION

Course:
Networked Control

Part I: Course introduction

14/09

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Outline

- Organization of the course
- Goals and topics of the course

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Organization of the course:

The course consists of about

- **About 48/50 hours (5 CFU)**
  - About 32 hours will cover theoretical subjects.
  - About 6 hours will cover numerical exercizes.
  - About 8/10 hours will be devoted to computer sessions.
- **Theoretical lectures:**
  - **Board** (although some introductory lectures will be delivered with the slides)**.**
  - There is no «commercial» textbook. **Lecture notes are available on webeep**.
- **Numerical exercises** on simple (e.g., scalar) examples.
- **Computer sessions with Matlab**.
  - Analysis and design of decentralized, distributed, and networked control systems.
  - You need your **laptop** with you when required.
  - **Matlab** must be installed.
  - You will also be required to download additional **free software** from the web (e.g., Yalmip).
- Slides, lecture notes, past exams with solutions, and related references will be made available to registered students through WeBeep, https://webeep.polimi.it **.**
- We have 4 hours a week:
  - Tuesdays, 12:15-14:15, room B.5.1.
  - Thursdays,12:15-14:15, room B.5.1.
- The course (google) calendar is available at the link
  - ➢ https://calendar.google.com/calendar/u/0?cid=NDc0ZHNybThzOGpmYWViYXZwMGwydW5uZnNAZ3JvdXAuY2FsZW5kYXIuZ29vZ2xlLmNvbQ (**most likely option**)
  - ➢ https://calendar.google.com/calendar/u/0?cid=djRpcXZtYzNiNmlxN28zbzEwZzhtbHBmaG9AZ3JvdXAuY2FsZW5kYXIuZ29vZ2xlLmNvbQ (**less likely option**)

Organization of the course:

All lectures will be held by

Prof. Marcello Farina

E mail: marcello.farina@polimi.it
Phone: 02 2399 3599

No specific appointments time. I am very flexible: send me an e-mail to fix a meeting when possible.

Recordings will be made available of

- Theoretical lectures
- Exercize lectures
- NOT COMPUTER SESSION!

Note that, however, lectures are mostly meant to be delivered in presence.

Organization of the course:

The **course** consists of three parts:

preliminars over control engineering

I. Introduction and theoretical background

II. Decentralized and distributed control structures → PROJECT

III. Networked control systems (NCS) → WRITTEN EXAM

different arguments

The **exam** consists of two parts.

- A **project work** to be completed and presented – possibly in groups - in an **oral discussion about Part II**.
  - Project: design – using the tools studied during the course – of distributed and decentralized controllers for a selected benchmark example.
  - The knowledge of the students about the topics covered in Part II will be also evaluated through specific questions.

- A **written test** will be taken, that includes **numerical exercises and questions about Part III**.

Outline

- Organization of the course
- Goals and topics of the course

What is this course about?

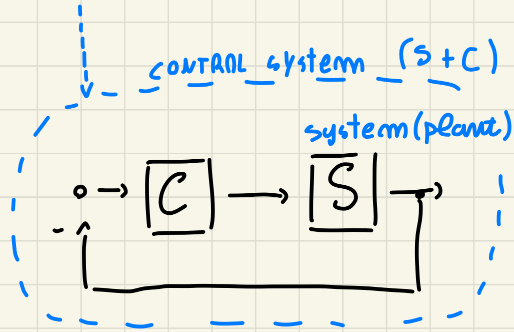
CONTROL

Issues arising in the **design** and **implementation** of **control systems** characterized by the presence of **components arranged in a networked configuration**

for component distributed

"CONTROL SYSTEM": both controller + system + all additional devices

things in between



you can consider different architectures, with more elements

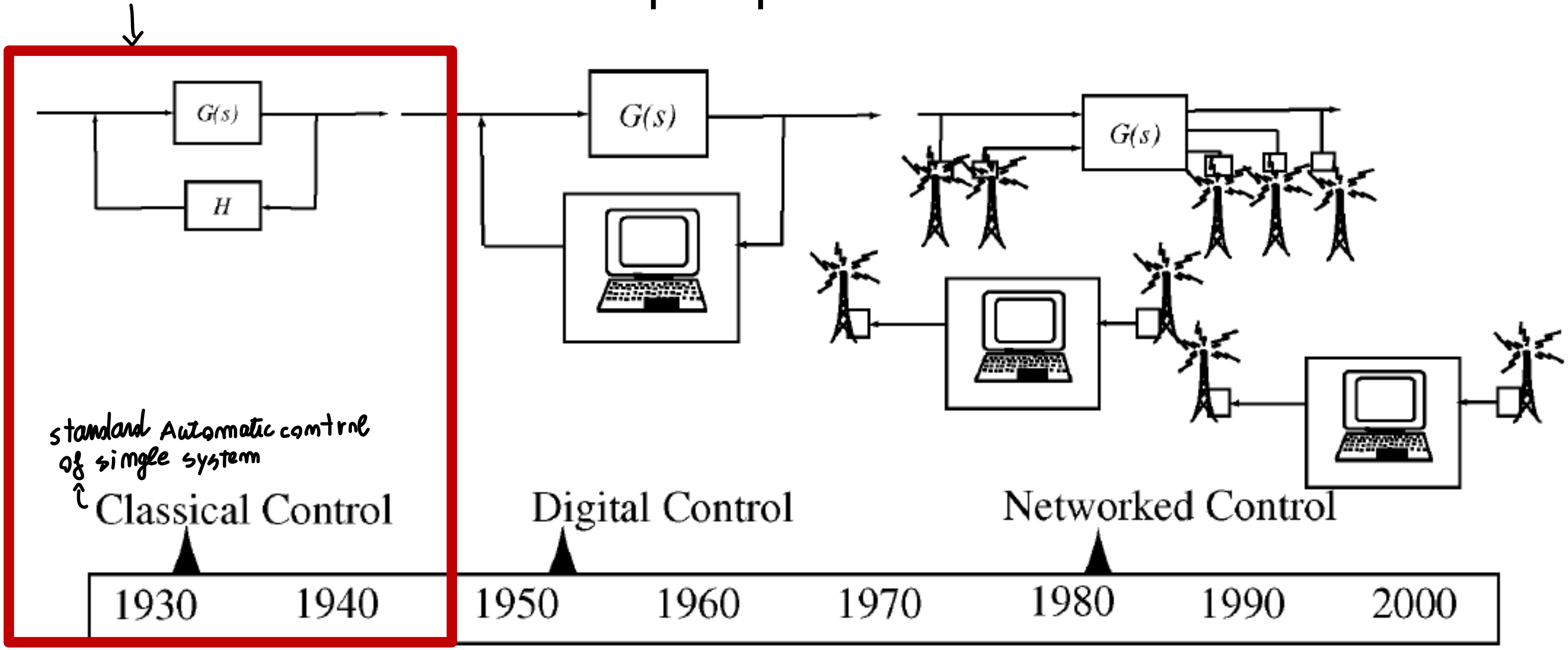
(we consider control system connected in networked configuration)

IT can be a big system of subsystem

OR many small systems that have to communicate (actuators, sensors...)

development of AUTOMATIC CLASSICAL CONTROL strategy for single system

Let us take an historical perspective



From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

Classical control problem

↓ usual problems, one system S with I/O representing internal state of the system → HOW TO CONTROL IT?

(intrinsically continuous)

Consider an *intrinsecally continuous-time* (possibly MIMO) plant $\mathcal{S}$

continuous-time system

$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_m(t) \end{bmatrix} \longrightarrow \boxed{\mathcal{S}} \longrightarrow y(t) = \begin{bmatrix} y_1(t) \\ \vdots \\ y_p(t) \end{bmatrix}$$

Control problem: how to make S behaves as we want!

- Steer the output vector $y(t)$ to a given set-point
- Stabilize the system/equilibrium point
- Optimality requirements (cost function)
- Convey robustness/ disturbance rejection properties
- ...

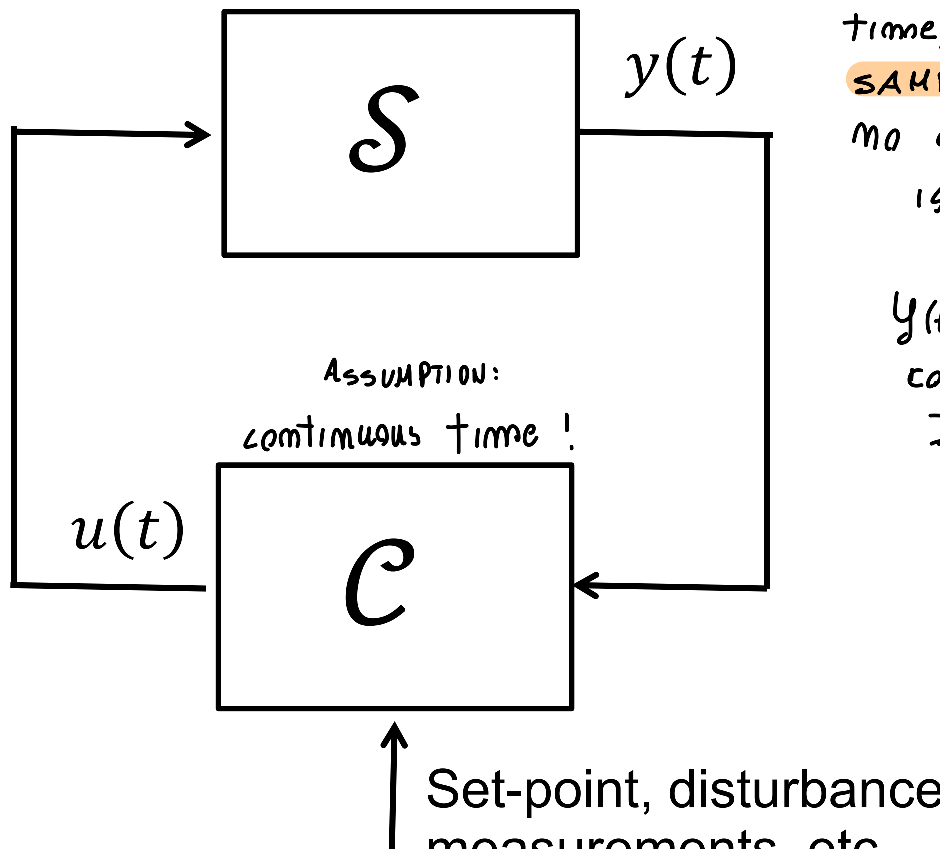
⇩

Design of the controller $\mathcal{C}$

Classical control problem

Overall feedback control system

FEATURE:
we assume controller C
in CONTINUOUS TIME ...
$u(t) \in \mathbb{R}$ cont
time, NO
SAMPLING, DISCRETIZATION
no quantization
issues...



$y(t) \rightarrow u(t)$
continuous
I/O

Classical control problem

continuous time C implemented by :

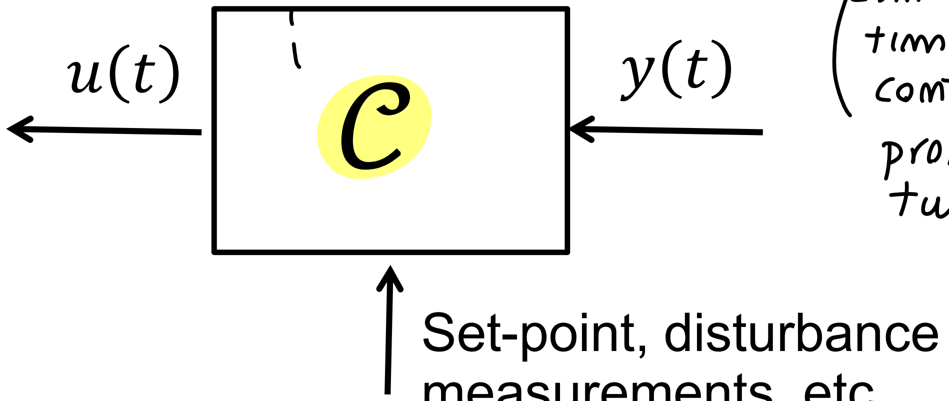
- From 1930 to 1950 solid theoretical foundations were established (mostly for SISO systems, frequency domain methods)
- Controller $\mathcal{C}$ was traditionally realized using
  - Analogue circuits
  - Mechanical systems
  - Hydraulic systems

⇒ continuous-time system

continuous time controller? problems on tuning, precision.. ↳ COMPUTERS →

T.F of regulator realized by technique continuous

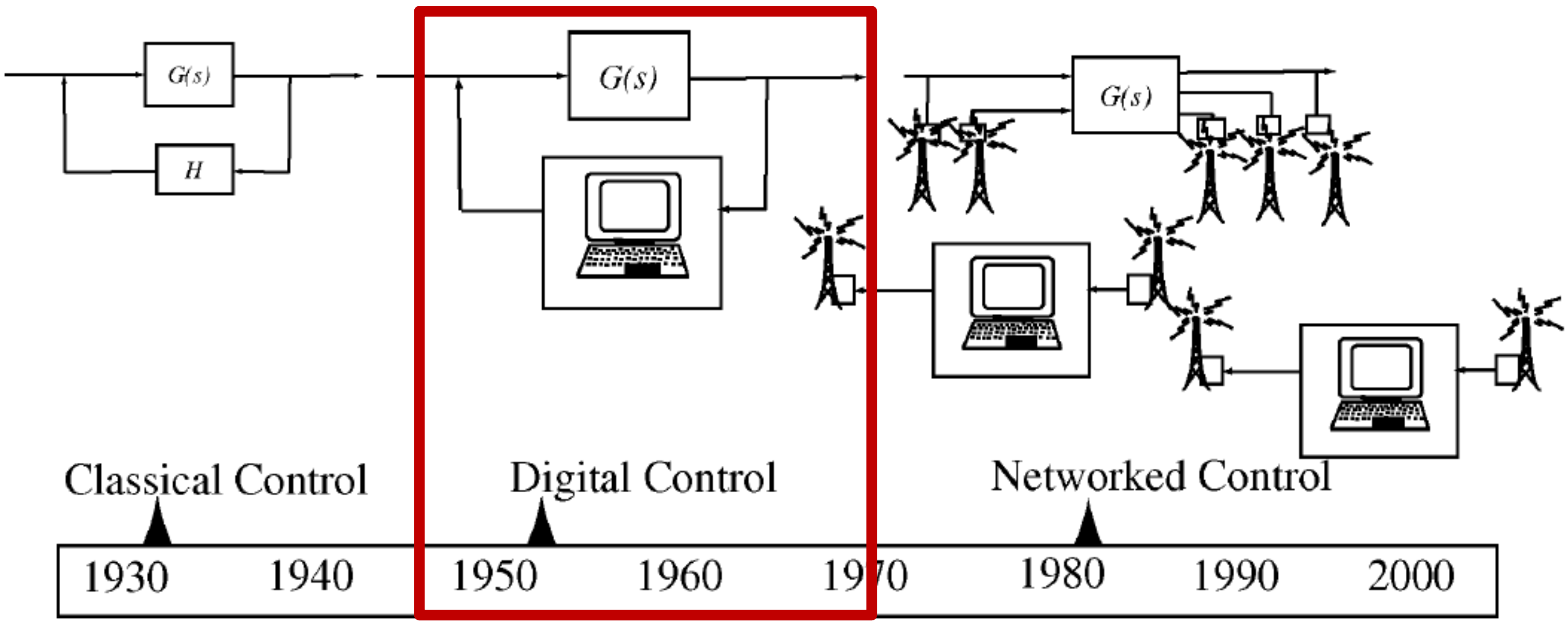
We have discrete systems, how to model continuous time ?



from the '50s
↓ use computer to implement control law

Historical perspective

↓ control inputs u(t) from computer!



From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

From classical to digital control

provide output according to a T.F designed, discretize by tustin ecc..

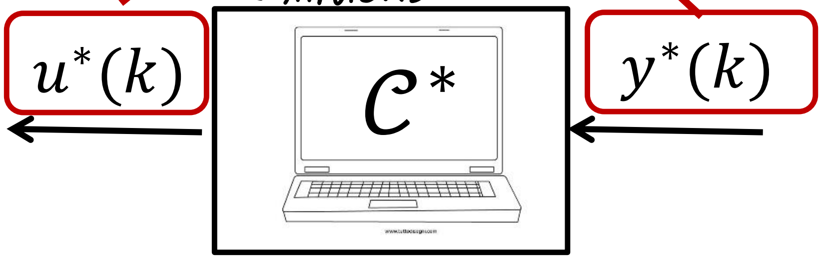
Starting from 1950 there has been a growing interest in the use of digital computers as instrumentation for feedback control

DIGITAL CONTROL:

Controller $\mathcal{C}^*$: *discrete-time system*

discrete-time signals := numbers used by computer

COMPUTERS



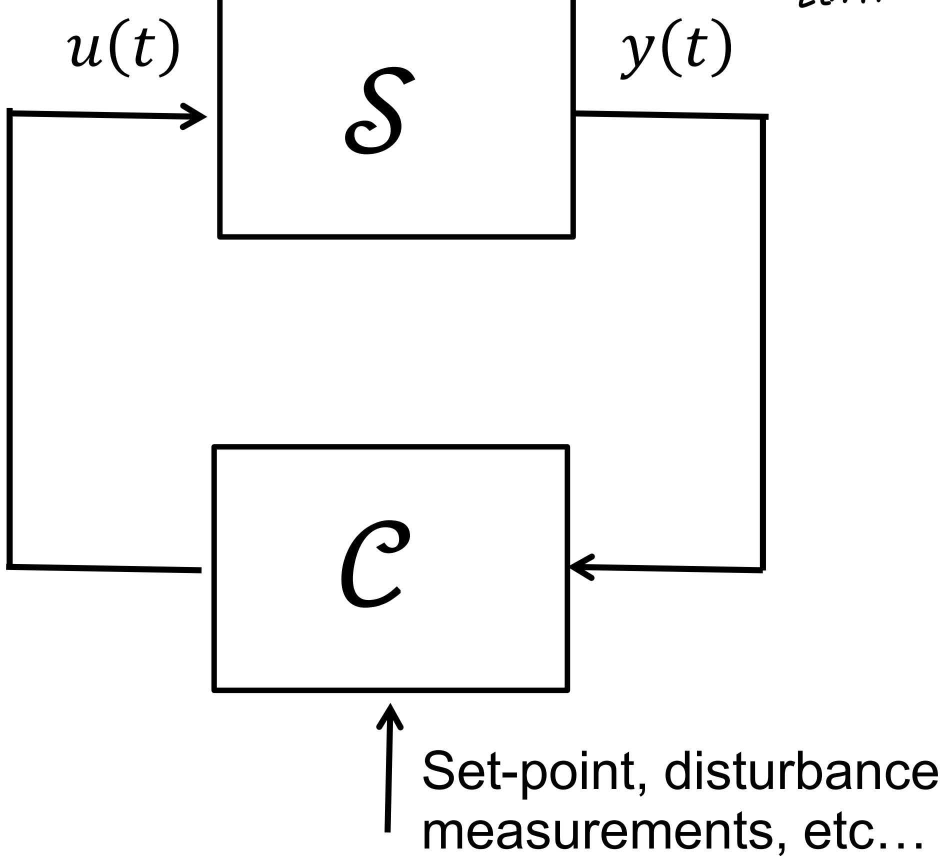
output according to designed T.F of controller... discrete time implementation

Set-point, disturbance measurements, etc…

Digital control

Overall feedback control system

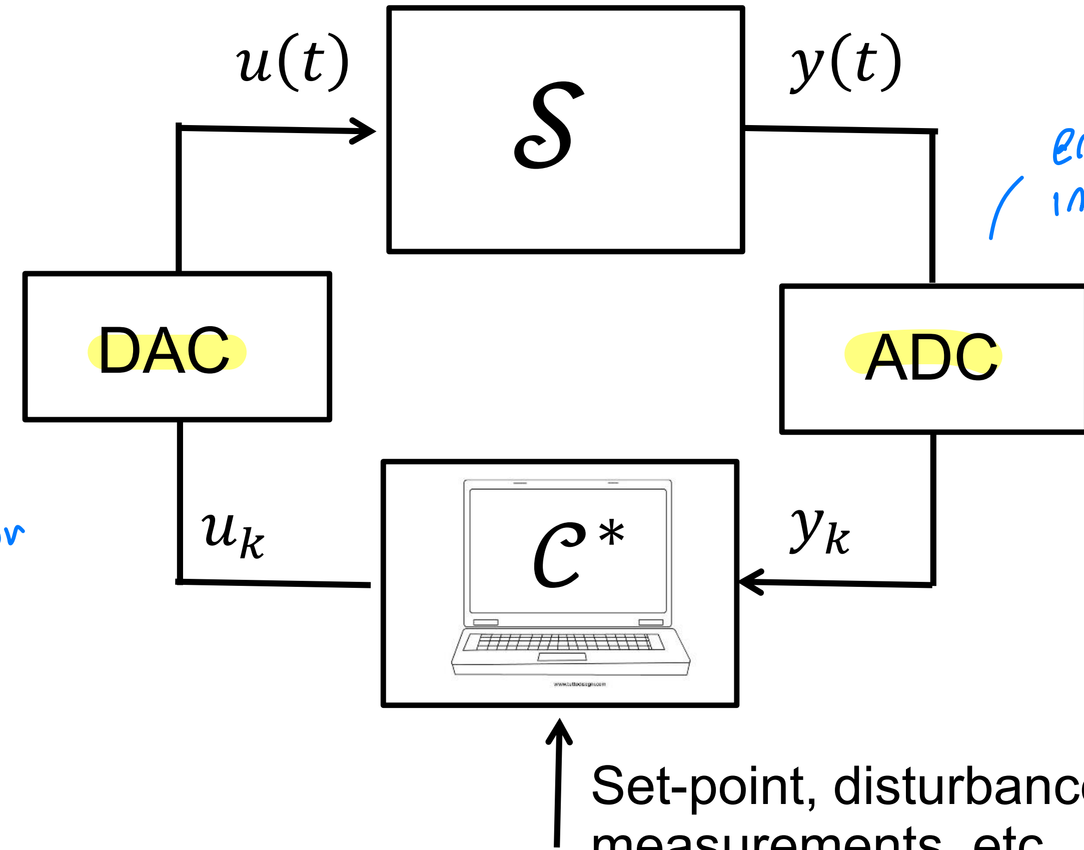
→ new configuration, converting ⇒



Digital control

Overall feedback control system

with proper converter
(abstract configuration..)



embedded in sensors

Digital-to-analog converter

embedded in actuator

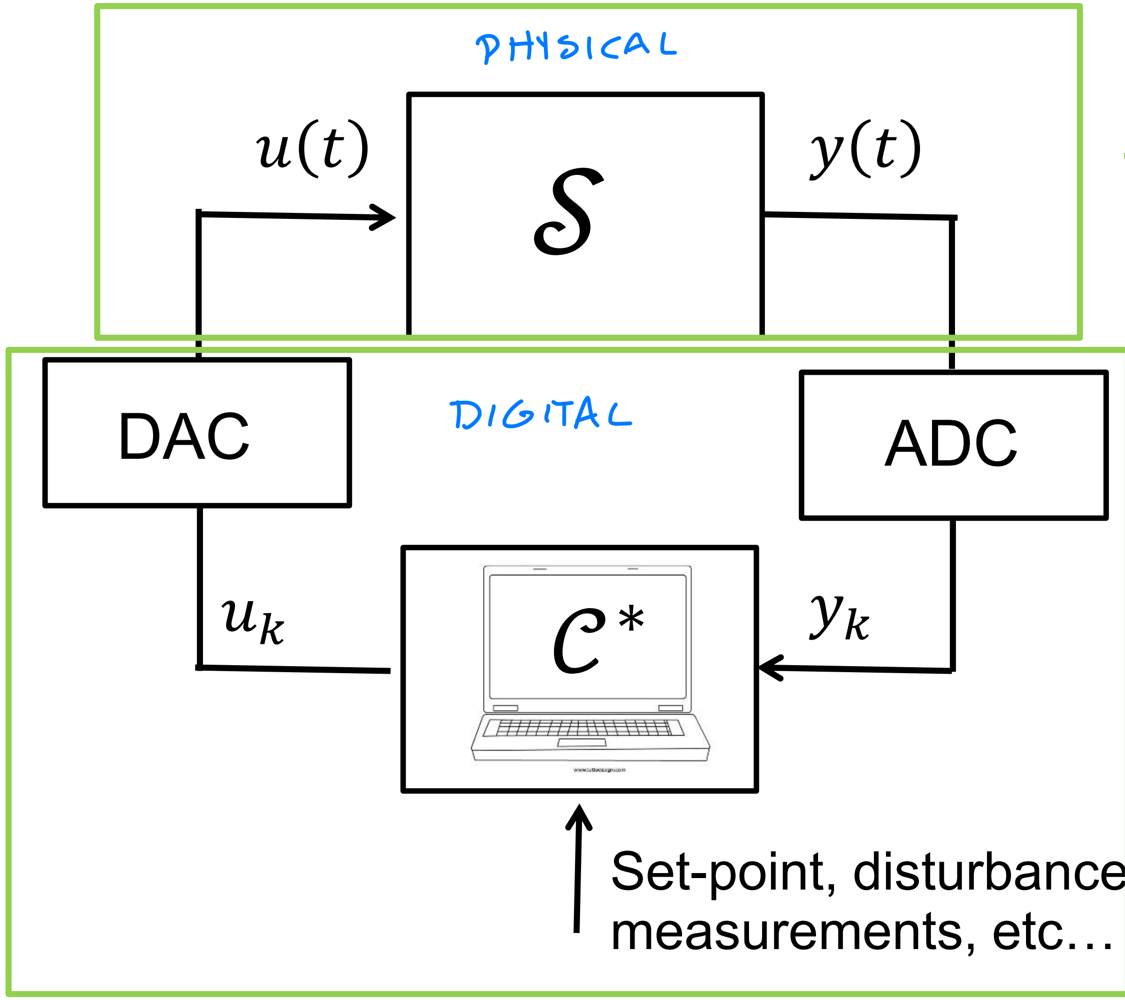
Analog-to-digital converter

Set-point, disturbance measurements, etc…

Digital control

DIGITAL CONTROL SYSTEM
(CYBER PHYSICAL SYSTEM) ← computer science name

Overall feedback control system



Physical plant (continuous-time)

Digital-to-analog converter

Analog-to-digital converter

Digital (cyber) controller

Assumptions

- NO significant delay, wire of information without loss (communications without delay.. time invariant)

  COMMUNICATION issue → loss of packet of INFO

  small delays negligible

- NO problem into control law implementation,

  ideal assumpt.. Ideal control law, system scale limited,

  easy to take into account and solve..

  control system without computational limit → small size of system

  + control action always computed without issues or delay

  due to large scale optimization or communication delays

Digital control

Issues: sampliny issue/ Discretization issue

- Effect of discretization/quantization of the signal sampled in time
  TAKE SAMPLE / ROUNDED, SAVED
- Choice of the sampling rate how to choice
- Choice of the word length
- Compensation of phase lags → related to sampliny, delay issue you need to compensate, deal with it
  sampliny brings delay

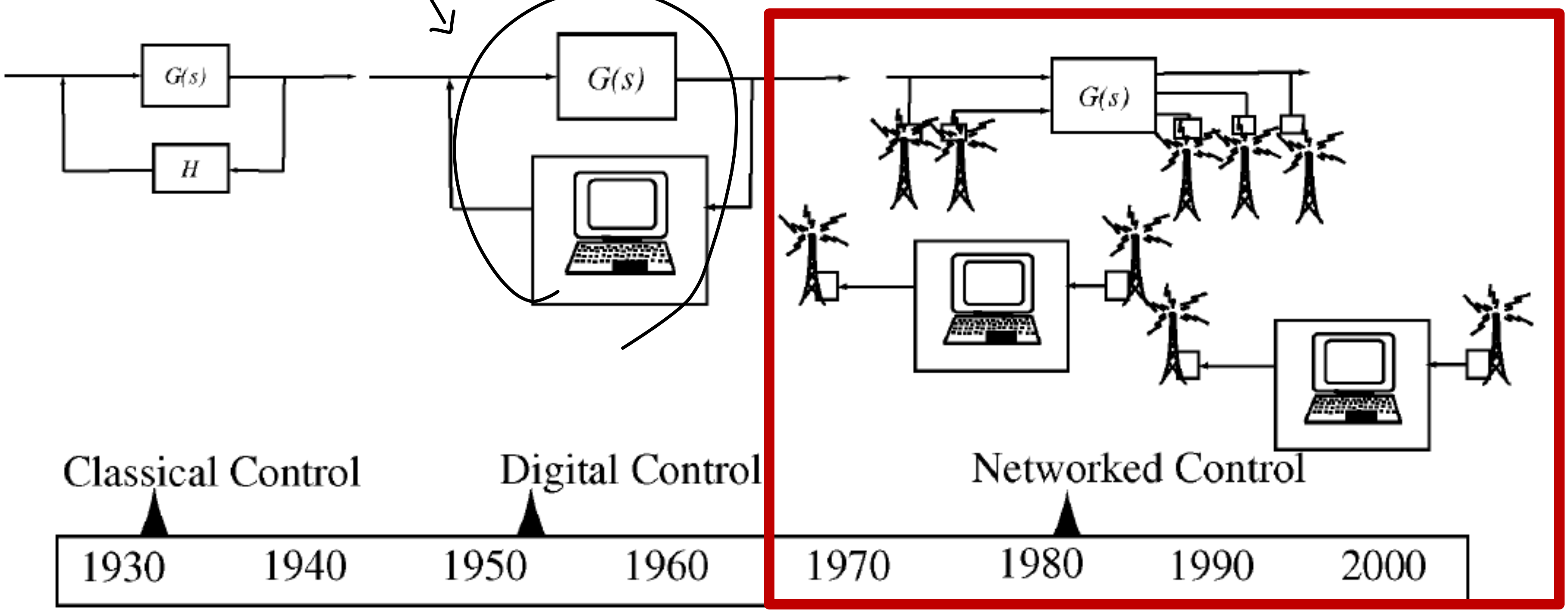
Digital control systems are nowadays also referred to as cyber-physical systems.
This terminology is especially used in the computer science area, where emphasis is given on:

- Software/hardware implementation
- Sensors
- Communication issues

up to now we discuss about classical control... with Assumptions!

Historical perspective

here you assume
NO significant delays (time invariant) ⟹ network control issue



From: J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28

From digital to networked control

Up to this point the main underlying assumption for application of (digital, possibly MIMO) control is that

the control system can be

- **designed** in practice **without computational limitations**, i.e., the size (scale) of the system does not prevent the control system to be designed. (NO limitations, small system)

- **applied without practical restrictions** and **without significant performance degradation** with respect to the nominal conditions.

↓ you can have large scale system and delays, communications which bring delay / loss of info

↳ NOW DAYS this problems are relevant

From digital to networked control

NETWORK control arrive later

NEW control problems! Large scale system to CONTROL

Since the 80's control applications have expanded, including plants and systems with **distributed components and/or with large scale.**

LESS ideal system: scale/communication issue

Systems with distributed components: systems where some parts are not co-located, geographically sparse, or having a modular structure (i.e., they can be regarded as a **network of different subsystems/components**).

Examples: many components which talk with eachother.. communication problem

- Airplanes/Cars/Motorcycles
- Chemical processes (LARGE SCALE)
- Electric networks
- Fleets of agents (autonomous vehicles on ground, air, and sea)
- Irrigation networks
- Manufacturing plants
- …

you can have large scale system with subsyst OR

system with lot of communicating pieces... control

Network communication with issues!

↓

less ideal system,

scale/communication issue

Examples

(PLANT): control a large scale GAS TREATMENT PLANT

CONTROL OVERALL PLANT?

2 SECTIONS

Natural gas treatment plant

(PLANT MODEL)

Natural gas:
Nitrogen
carbon dioxide
methane
ethane
propane
iso-butane
n-butane

Refrigeration section
(divide INPUTS into sets)

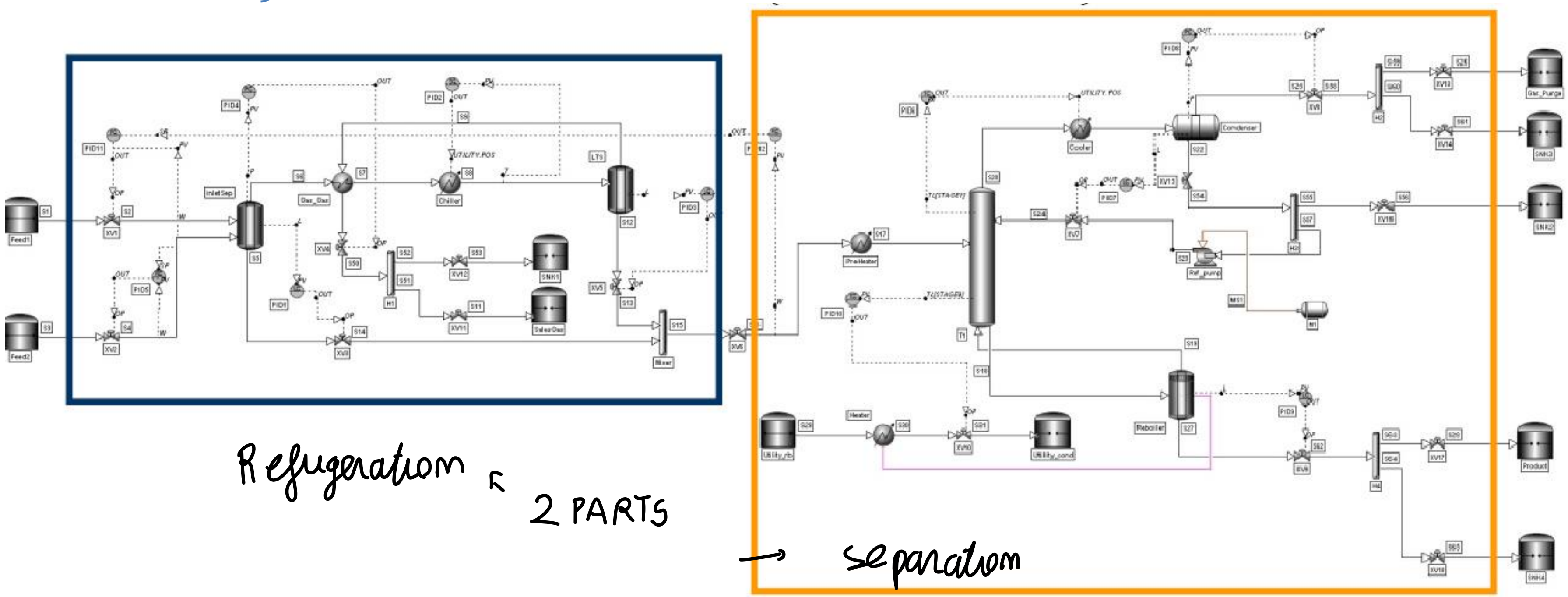
SET 1
Propane
iso-butane
n-butane

Separation section

SET 2
Nitrogen
carbon dioxide
methane
ethane

Pipeline, gas flaring

LOST GAS



Refrigeration

2 PARTS

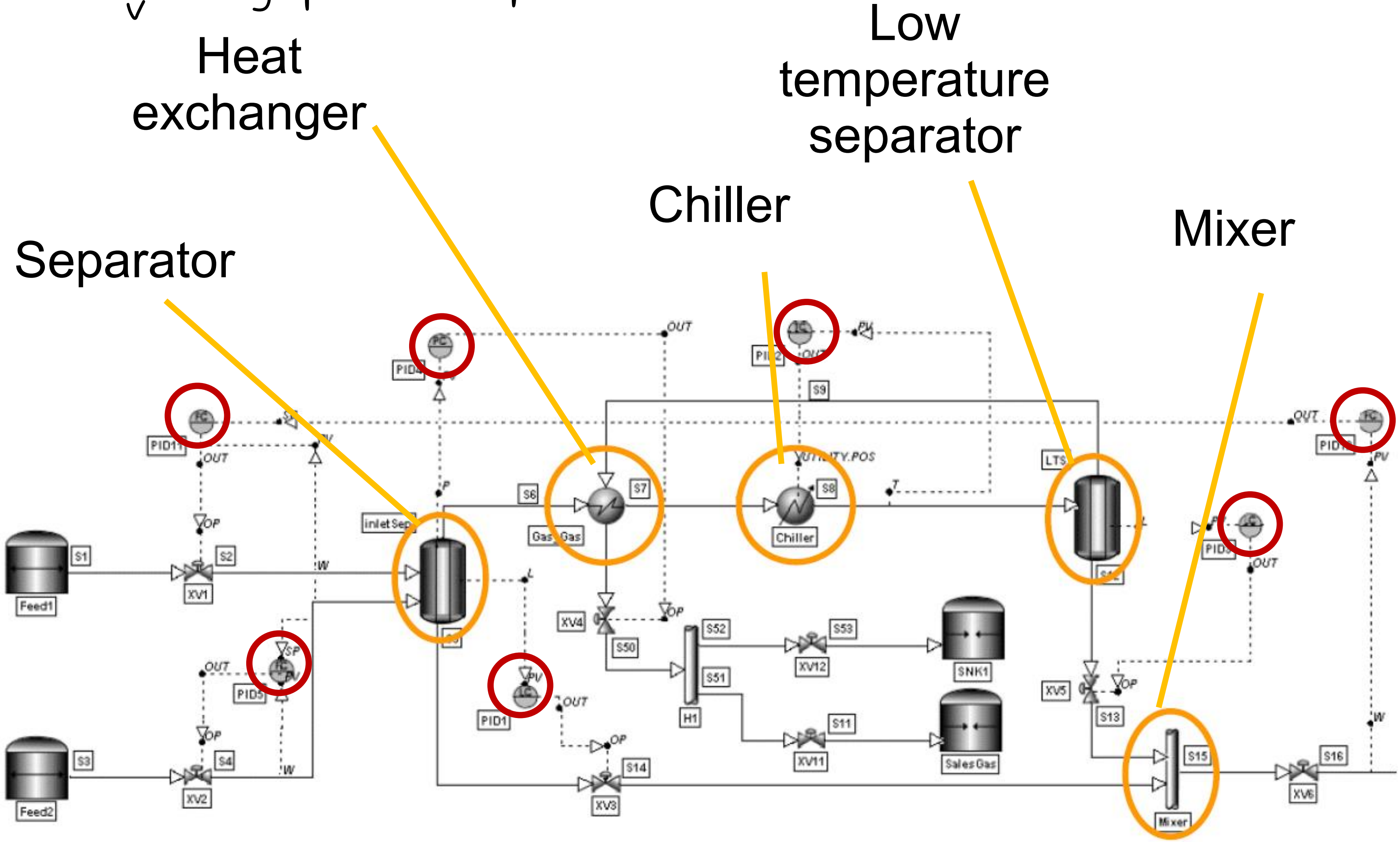
separation

Examples

First section

Natural gas treatment plant – refrigeration section

many part compose it... each part as complex system



← we can control with single PID ∀ sub system...

effective CONTROL BUT NOT optimal

↓

for optimality, we cannot discard interaction

↑

NOWDAYS

a possible solution is control each subsystem separately with a PID for each one → effective BUT NOT OPTIMAL

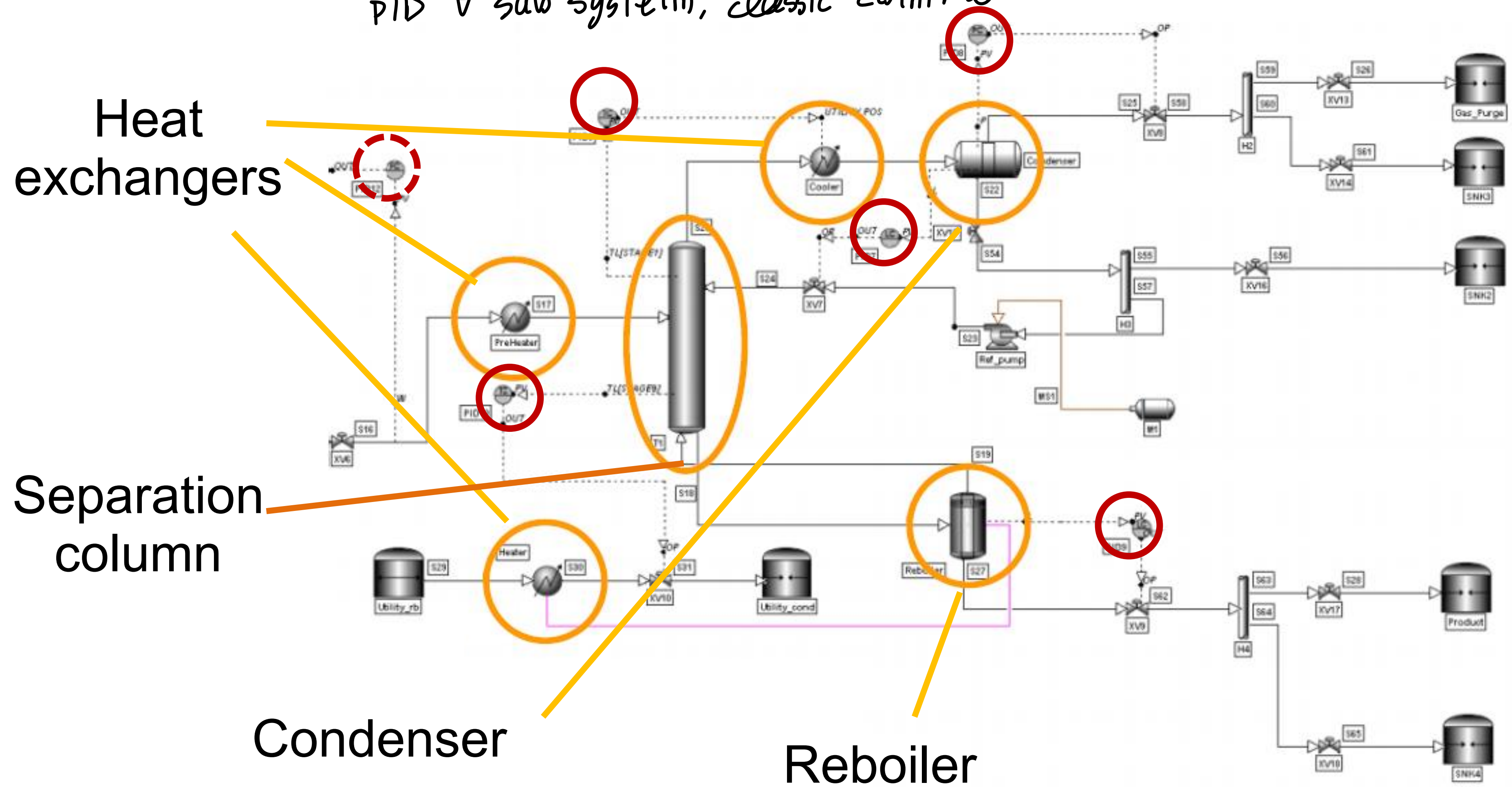
STRONG INTERACTION! in reality..

Examples

PID controller designed separately… problem limitations! still…

Natural gas treatment plant – separation section

PID ∀ sub system, classic control



Examples

Plant model!

many subsystems

Natural gas treatment plant

each subsystem is like a PLANT!

(PROBLEM)

- **Large-scale system**, composed of many interacting subsystems
  - Subsystems are geographically distributed
  - Each subsystem must be subject to control (is itself a complex and critical system)
  - Interactions must be considered for optimal performances ← CONTROLLER considering interactions

MANY complex subsystem... how to CONTROL overall?

- MIMO system: (**centralized**) multivariable control?

consider as a unique system, COMPLEX! ← approach working IN THEORY!

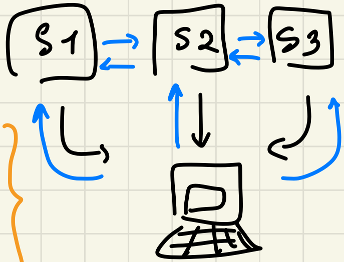
- Challenges:
  - Large control problem dimension – **computational challenge**
  - Can information must be conveyed to the control station efficiently, «rapidly», and «simultaneously» from the different sensors and to the different actuators? **Information structure constraints** may be present!

DISTRIBUTED can be best choice...

- **Present control scheme**: each subsystem is controlled independently (**decentralized** control): loss of performance/risk of instability.

← considering all the system as MIMO make the problem HARDER... how to CONTROL that?!

sub systems :



you need communication NETWORK
instantaneus INFO to computer

you need INFORMATION coming instantaneusly and synchronous

ISSUES
- SAFETY
- ROBUSTNESS
- COMUNICATION
- COMPUTATIONAL

simgle computer which has to take care of all, communications

takes time.. ←

↓

ALTERNATIVE :
control separately → Discard interaction controller embedded in each SUBSYSTEM.. NO communication issue, NO computational issue...
NOT big issues as before

each controller for each sub system ←

2 APPROACH

CONTRO
centralized : NOT so Relyble (one control for all)

MIDDLE..

+ DISTRIBUTED (middle approach): account interaction exchanging some data communicate
HELP
(take into account interactions) ↓ provides each sub syst with local controllers + communication

CONTROL
Decentralized : All controlled by local loops.. but Done in a naive way → Neglect interactions

Examples

collect data underwater: measurement units to take data of the water → quality of see ecc.. (moove in some formation network of bots underwater)

Fleets of underwater autonomous vehicles for sensing

Goal: **design a** mobile sensor **network** to collect measurements from a field

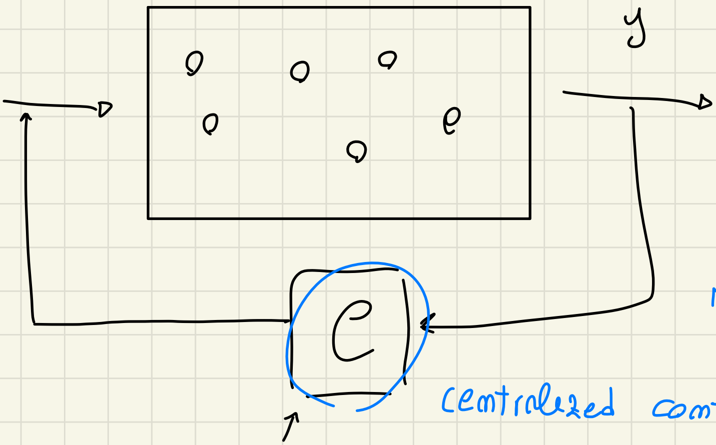
- e.g., salinity, temperature, chlorophyll fluorescence (i.e., a proxy for the concentration of phyloplancton, bottom of marine food chain, indicator of the state of the marine ecosystem and of the atmosphere)
- optimal measurements: sensors should **move in a rigid formation.**

how to control a fleet of gliders, to provide effective measurements and mantain stable behaviour, travel effectively to travel along wide area

Moving sensors are underwater gliders:

- possibly autonomous underwater (e.g., for ocean depths) vehicles (AUV), equipped with sensors for measuring the environment.
- can collect data continuously over periods of weeks or even months.

From: N. E. Leonard *et al*. Collective motion, sensor networks, and ocean sampling. *Proceedings of the IEEE*, 95 (1), pp. 48-74 to CONTROL you could consider the overall fleet of gliders as OVERALL SYSTEM, to control



assume to close the loop in this way

centralized control ?

- NO Flexibility wrt failures
- adopt plug & play scenario when some gliders can be OFF, added...
  NOT doable in CENTRALIZED where I/O is fixed!

optimal ? but unfeasible!

you don't consider failure issues!

↳ and you can have lots of gliders..

BIG # state variable -> complexity raise

Examples

Fleets of underwater autonomous vehicles for sensing

It is not possible/convenient to consider the **fleet as a single system** to be controlled in a unified (centralized) way:

- **Lack of flexibility** (i.e., with respect with the number of agents).
- **Unreliable** (i.e., with respect to faults at single agent level).
- Possibly very large-scale system (when the number of gliders is large): **computational limitations**: control actions must be taken at a centralized level for all the agents. too many variables.. computational complexity of problems increase
- Severe **communication limitations**: measurements are taken at a local level, they must be transmitted (istantaneaously and sinchronously) to the central control station and control actions must be transmitted (istantaneously and sinchronously) from the control station to the local actuators.
  - Radio frequency communication (used in other contexts, e.g., land) cannot be used in water.
  - Acoustic communication not practical (distances of kms between gliders and with the central station). OPTIMAL CONTROL? hard ⇒

when underwater communication miss...

SOLUTION: hyerarchical approach, 2 level control... → LOW CONTROL UPPER CONTROL LAYES

Examples

Fleets of underwater autonomous vehicles for sensing

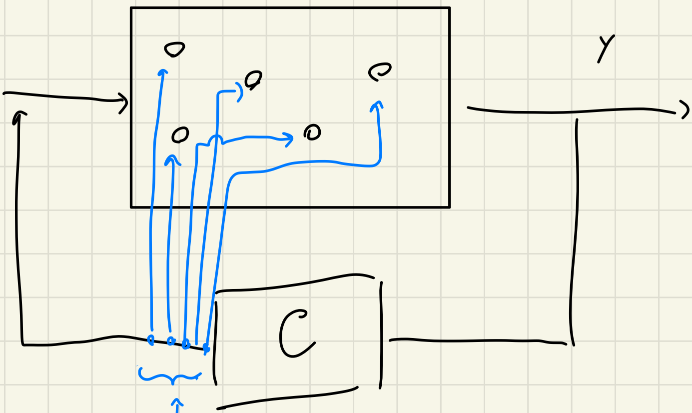
provides with SLOW RATE: high sampling time info to the single gliders where it has to go

Adopted solution: 2-level control

centralized := taking care NOT of Inputs

- **HIGH LEVEL cooperative control** (implemented in a **single control station**, but possibly also in a **distributed** way):
  - Definition of way-points (i.e., trajectories) using (distributed) artificial potentials, to maintain a formation (virtual body).
  - This allows to **stabilize the desired formation**.
  - **Communication** is available only when gliders surface (periodically and asynchronously), when also
    - GPS measurements are obtained and transmitted to the control station.
    - waypoints are transmitted by the control station to the gliders.
  - **Chellenges** (tradeoff):
    - High frequency surfacing for better coordination
    - Low-frequency surfacing for saving energy

- **LOW LEVEL** (local):
  - Heading and pitch control
  - Follow way-points
  - Necessary data collected and estimated locally



hyerarchical,
don't talk to
each other, they
talk with
a single C
controller

(INFORMATION
to each
glidar)
↓
waypoints
to follow, HIGH (HIERCHICAL)
LEVEL don't control the system,
but give info on where to go

communication with Gliders
in surface, NOT underwater!

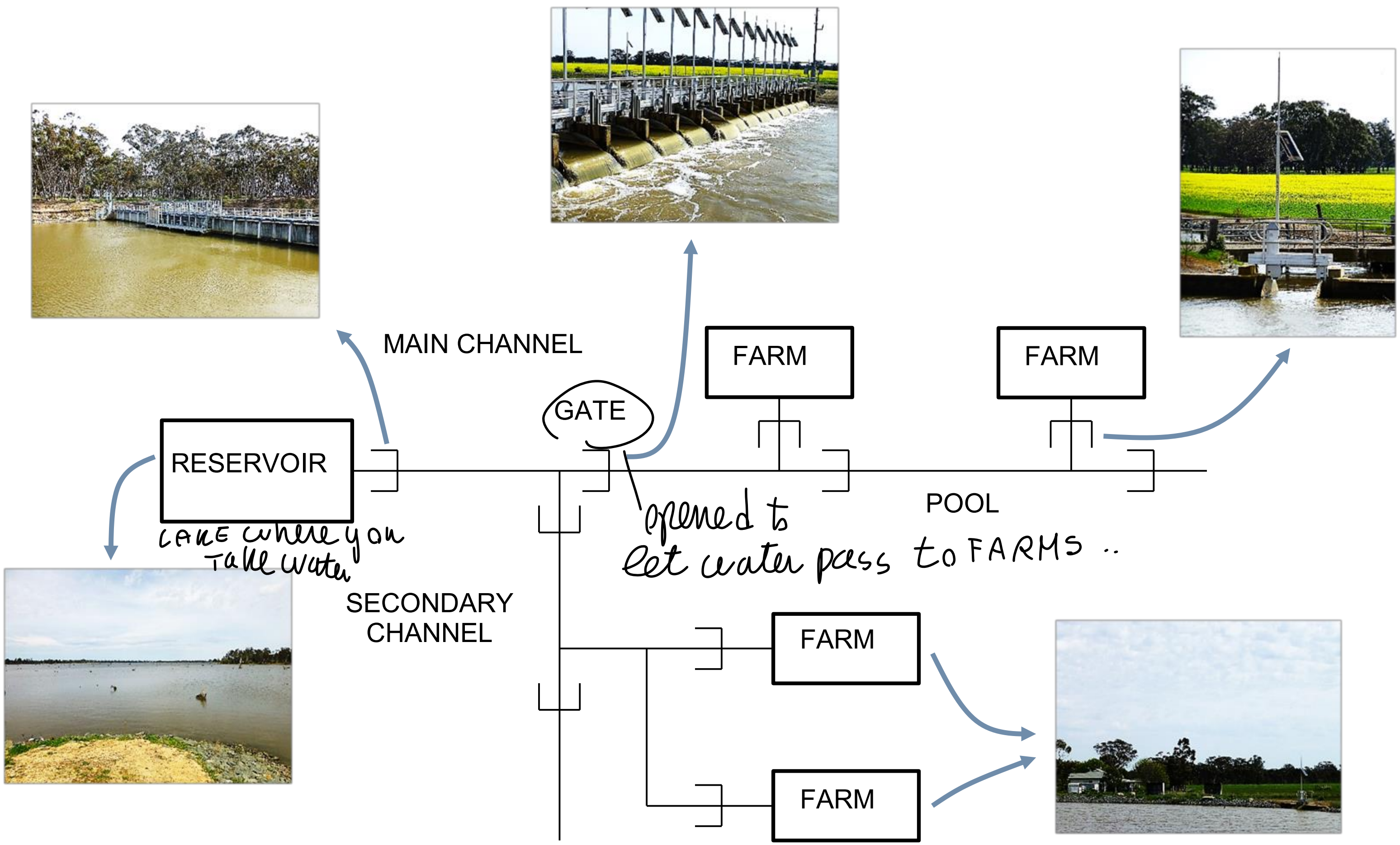
LOW LEVEL take care of single Input, don't
(communicate.. talk with central one (DECENTRALIZED)
→done locally, based on waypoints

in surface GLIDARS provide position to HIGH LEVEL controller
(hyerarchical) which compute NEW waypoints

Examples

large scale control of irrigation channels

Irrigation networks



From: M. Cantoni *et al*. Control of large-scale irrigation networks. *Proceedings of the IEEE*, 95 (1), pp. 75-91

FARMS: Nowdays manual actuation... delays, probleems errors supply (disturbances ON NETWORK)

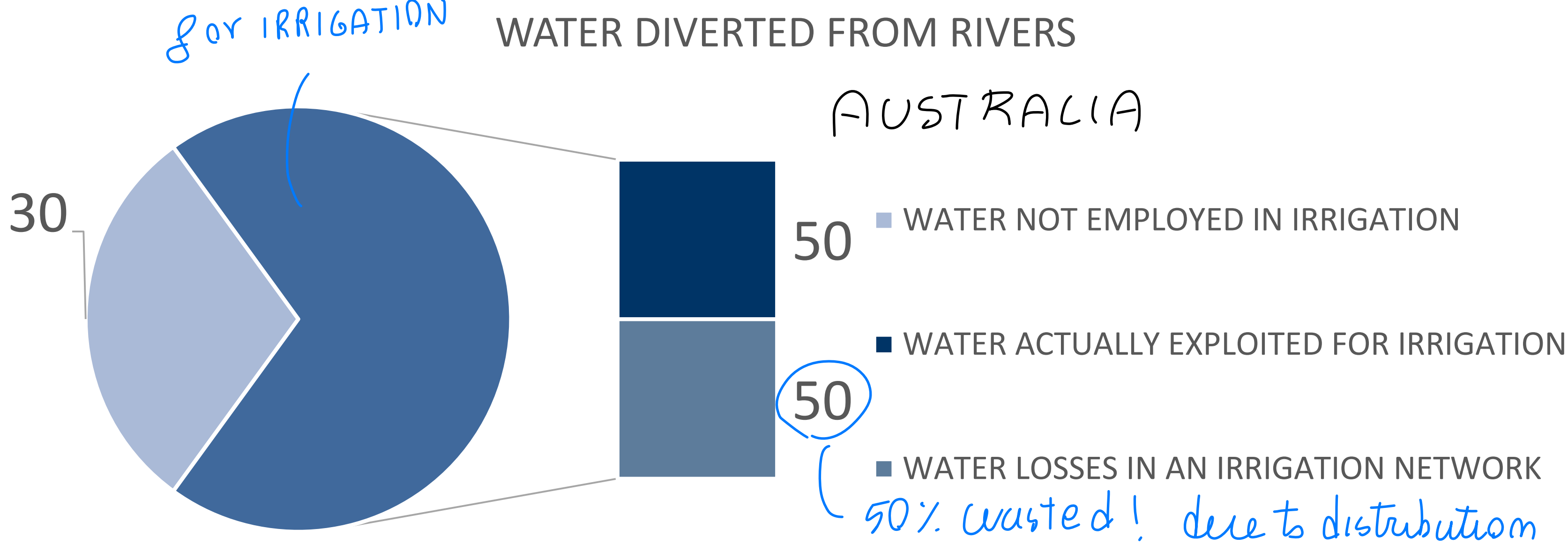
↳ How to automate the system..

Examples

Irrigation networks

70% used for IRRIGATION

AUSTRALIA



50% wasted! due to distribution issues! OVERFLOW of water OR NOT enough ...

Large scale distribution losses

- water over-supply
- water under-supply

Distribution losses at farms

- Manual actuaction of the gates

A fully automated control can improve water distribution efficiency up to 85%

Aquatic ecosystem emergency

Overflowing and water wastage

Errors and delays in water supply

Examples

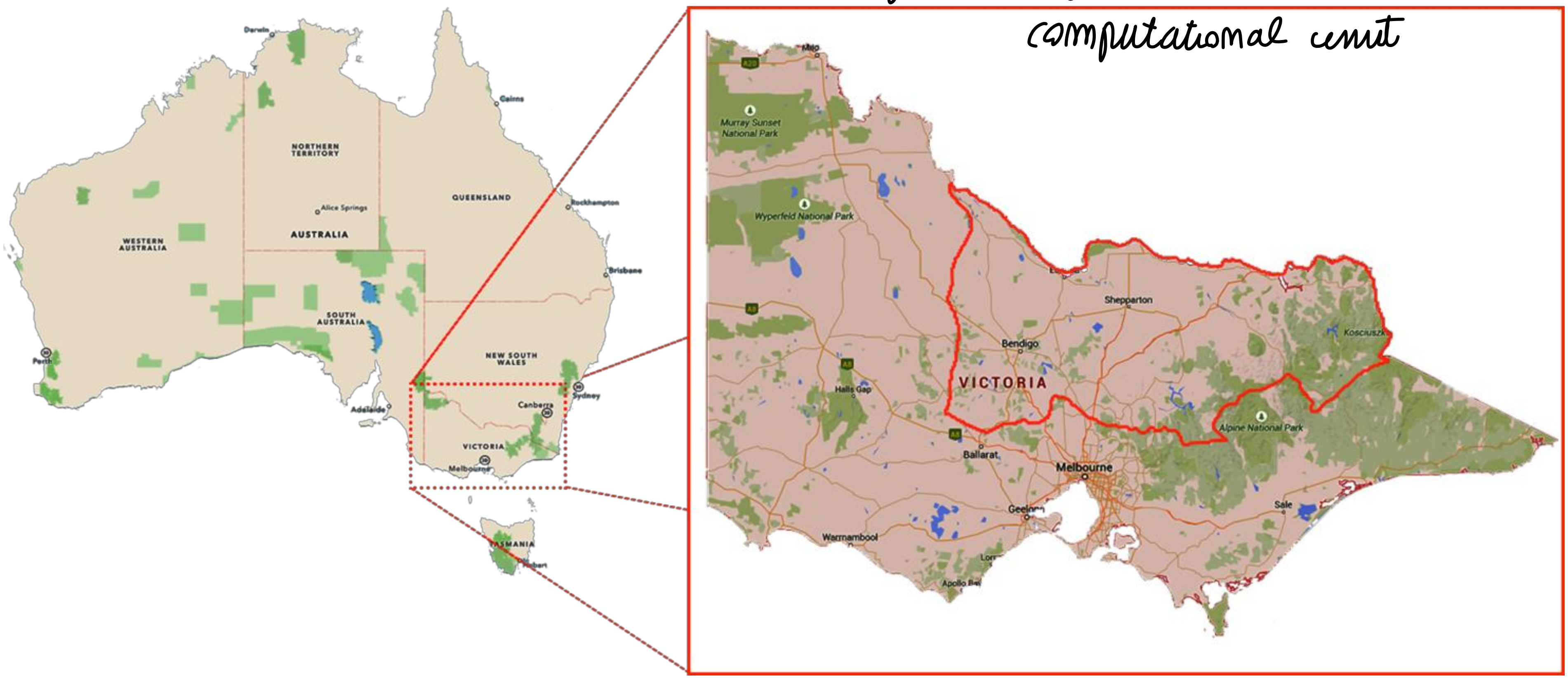
Irrigation networks

Where ? ↘ VERY HUGE area

- Murray-Goulburn Irrigation district:
  - 7000 Kilometers of channels
  - 68000 km^2 area
  - 21000 farm outlets

|| → LARGE-SCALE ! control issue, system model very huge, unfeasible with a single computational unit

⇒ **Large-scale system**



Examples

↓ possible model sketch

Irrigation networks

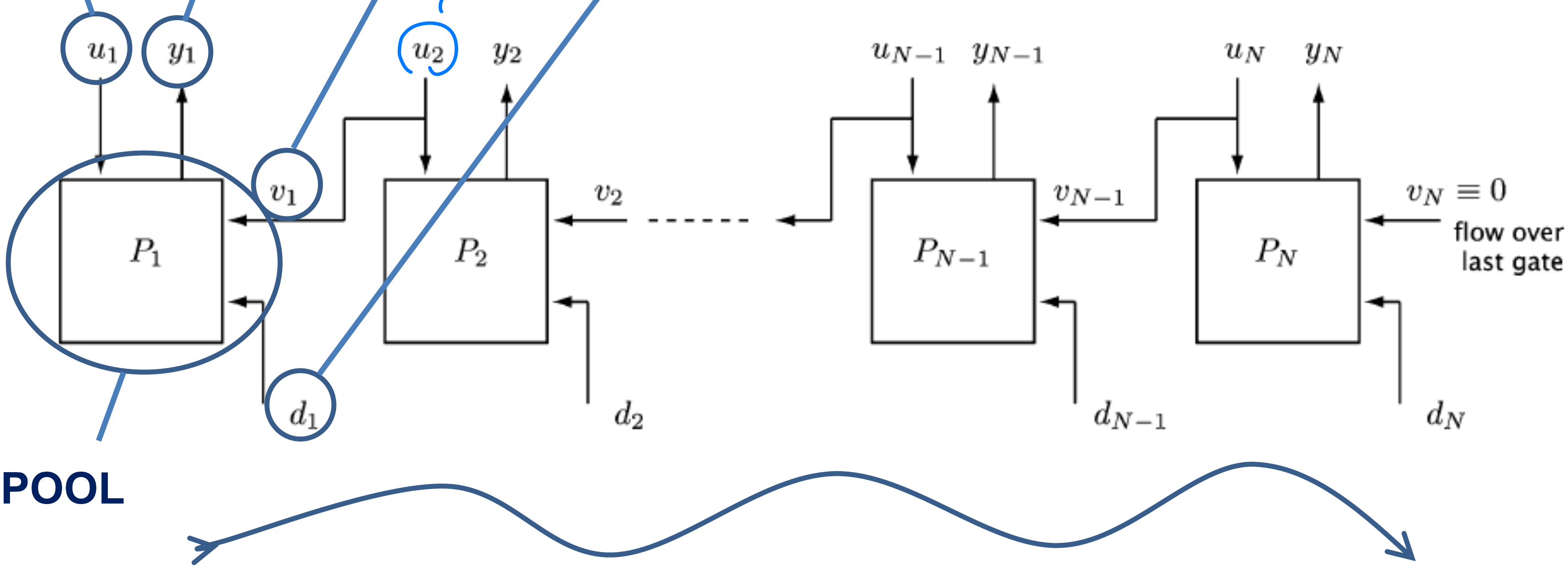
Water level in the pool

Flow from the pool

Gate opening

+ Disturbance : related to leakage, rain, farm operation..

input affecting two pools, control GATE between P1, P2



POOL

Water flows downstream

Examples

think about CENTRALIZED, IF you can deal with that model

Irrigation networks

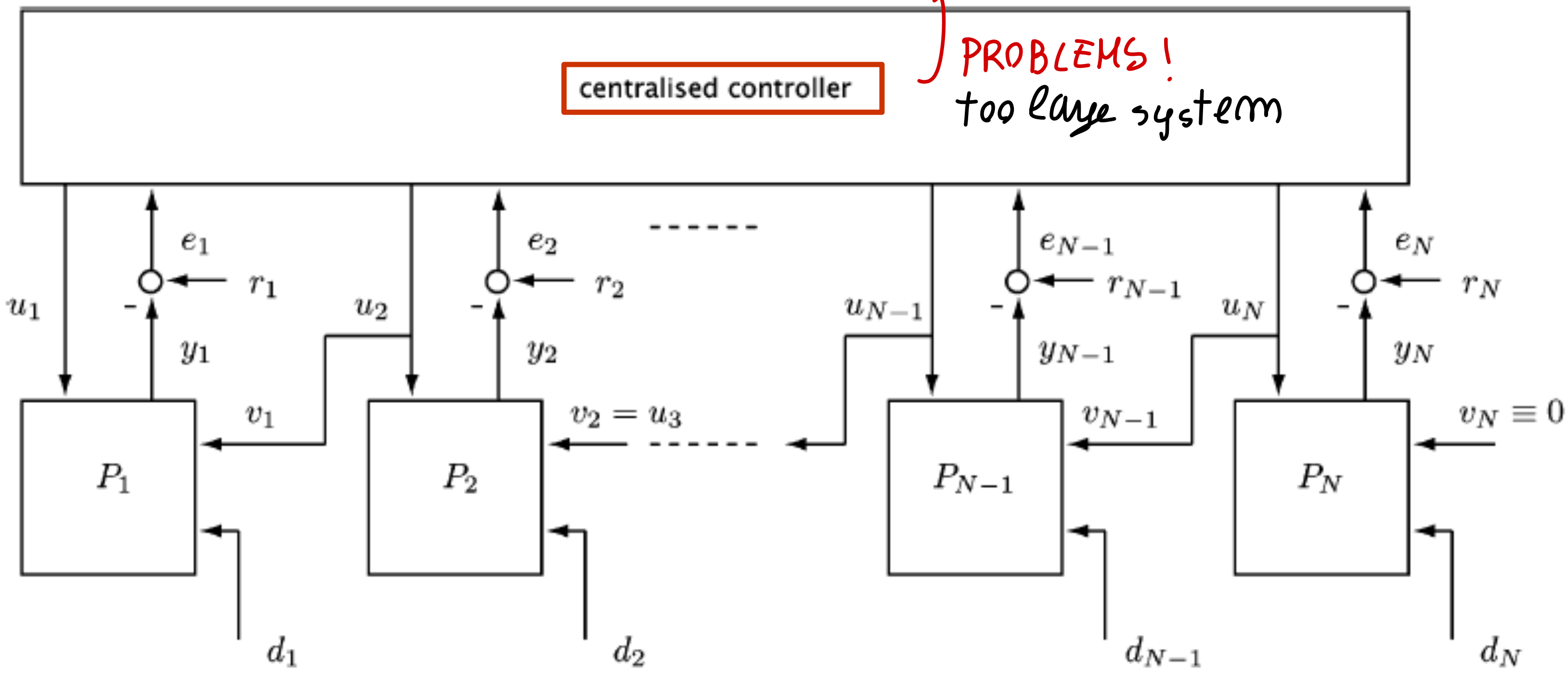
↓ theoretical solution :

- Centralized computation
- Centralized information structure

Not compatible with a possibly very large-scale system!

(COMUNICATION/ FLEXIBILITY.. ISSUE)

Distributed or decentralized solutions are necessary!



Examples

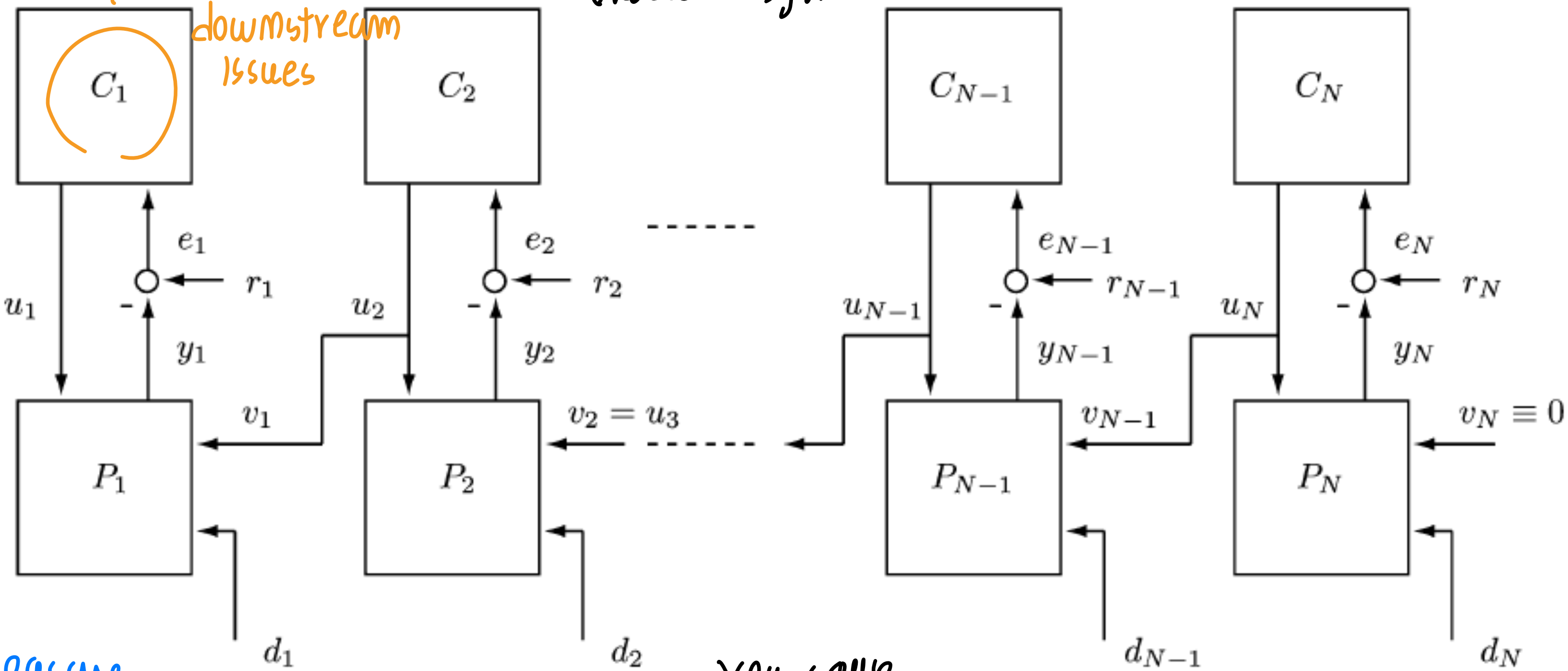
Irrigation networks

- Local computation
- Decentralized information structure

Actions, taken at local pool levels, are computed locally based on local information

control single part.. BUT you have string instability, d_i distribute overall system

NOT aware of downstream issues



you care downstream problems

measure water and control single GATES → BUT STRING instability, disturbances propagate downstream!

Examples

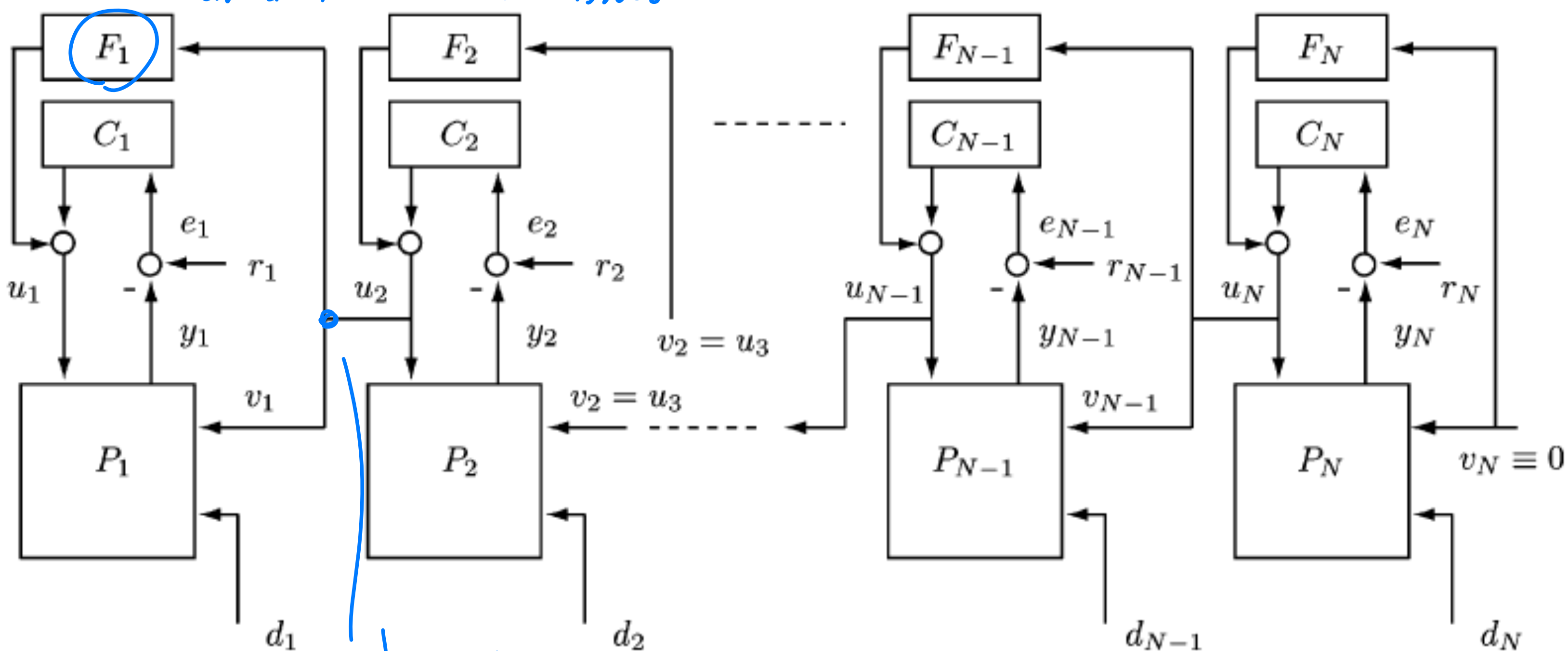
BETTER SOLUTION

- Local computation
- Distributed information structure

↓ communications necessary to avoid downstream issues

Irrigation networks

Actions, taken at local pool levels, are computed locally based on local information + information transmitted by neighboring gates' control systems



downstream communication to avoid issues

Effective & Feasible approach!

↳ take care of single GATE + compensator is easy!

Examples

(previously we had LOGICAL interactions in between) ← NOT LARGE scale system... NOT big system and lots of parts...

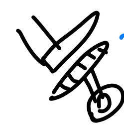
Control systems in cars

Present-day cars may be characterized by a huge number of

- **Sensors**:

each element of a car is a sensor giving info to engine CONTROL system (not directly mechanically)

  - **Pedals** (accelerator, brake)
  - Steering wheel sensor
  - Encoders: four wheels angular velocities
  - Inertial measurement unit (IMU): 3D accelerations + pitch, roll, and yaw rates
  - Pressure in the braking system
  - Suspensions length
  - State of all the actuators (throttle valve, air flowrate in the engine)
  - Lambda sensor: air/fuel ratio
  - Emission measurements
  - Wheel pressure
  - Cabin temperature sensors
  - …



PEDAL → INFO → SYSTEM CONTROL ENGINE

↑ here you have CAR CONTROL system, small space BUT lots of different sensors!

differently from GLIDARS, with common goal
↓

Examples

give INFO to steering control system

Control systems in cars

Present-day cars may be characterized by a huge number of

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    - Pedals (accelerator, brake)
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    - Lambda sensor: air/fuel ratio
    - Emission measurements
    - Wheel pressure
    - Cabin temperature sensors
    - …

Examples

Control systems in cars

Present-day cars may be characterized by a huge number of

- **Sensors**:
  - Pedals (accelerator, brake)
  - Steering wheel sensor
  - **Encoders**: four wheels angular velocities
  - Inertial measurement unit (IMU): 3D accelerations + pitch, roll, and yaw rates
  - Pressure in the braking system
  - Suspensions length
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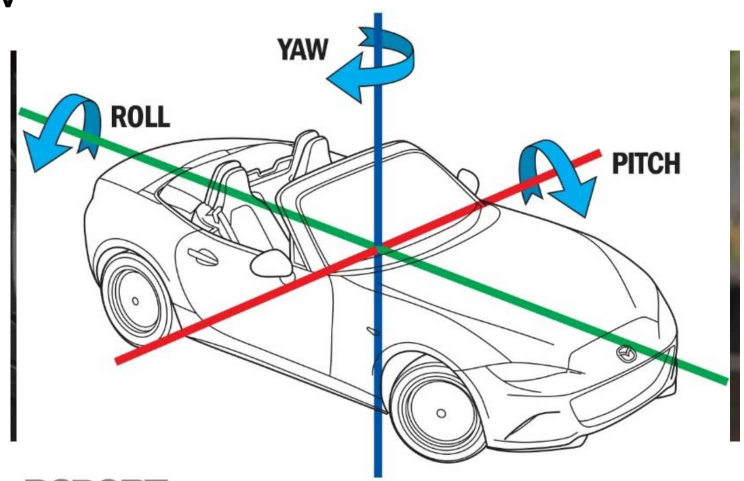

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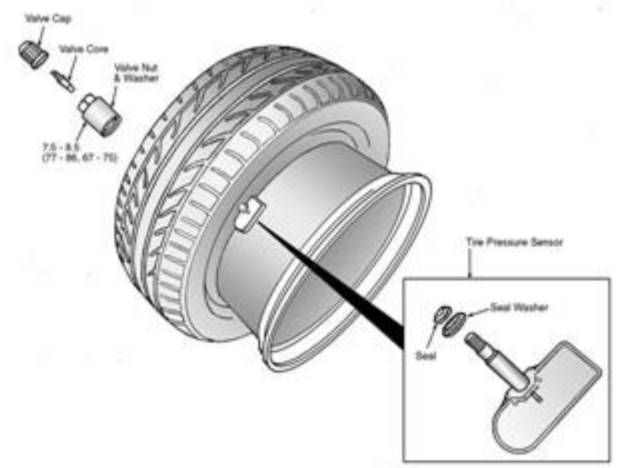

Examples

even a large number of actuators

Control systems in cars

Present-day cars are characterized by a huge number of

- **Actuators**:
  - **Throttle valve**: air flow into the engine
  - Fuel injector
  - Hydraulic modulator (braking pressure actuators)
  - Semi-active suspensions
  - Air conditioning actuators
  - Brakes
  - …

Examples

Control systems in cars

lot of control units, each for a single task

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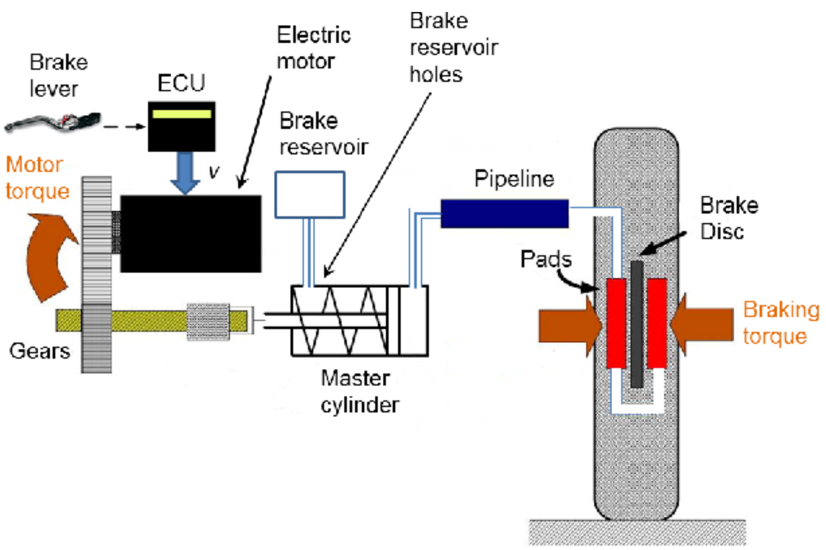
    - …



Examples

Control systems in cars

Present-day cars are characterized by a huge number of

- **Actuators**:
  - Throttle valve: air flow into the engine
  - Fuel injector
  - Hydraulic modulator (braking pressure actuators)
  - Semi-active suspensions
  - Air conditioning actuators
  - Brakes
  - …

- **Control systems**:
  - **ECU** (Engine Control Unit): to follow traction torque setpoints given by the acceleration pedal or other vehicle control systems, while controlling emissions. Talks with actuators: throttle valve, gas recycle, air injection.
  - **ABS** (anti-lock braking system): to follow deceleration setpoints given by the braking pedal without incurring in slips.
    - Reduce stopping distance.
    - Maintain Steerability
    - Maintain lateral stability.

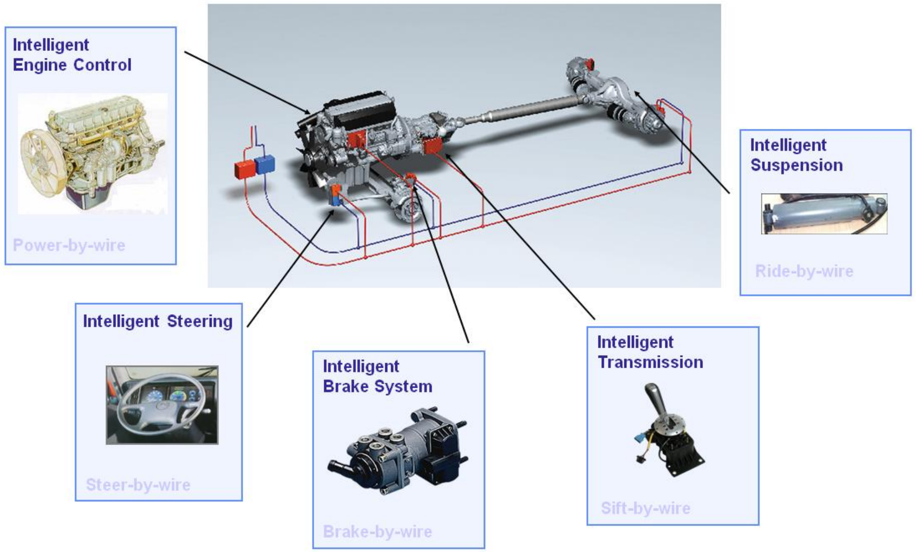
Examples

Control systems in cars

Present-day cars are characterized by a huge number of

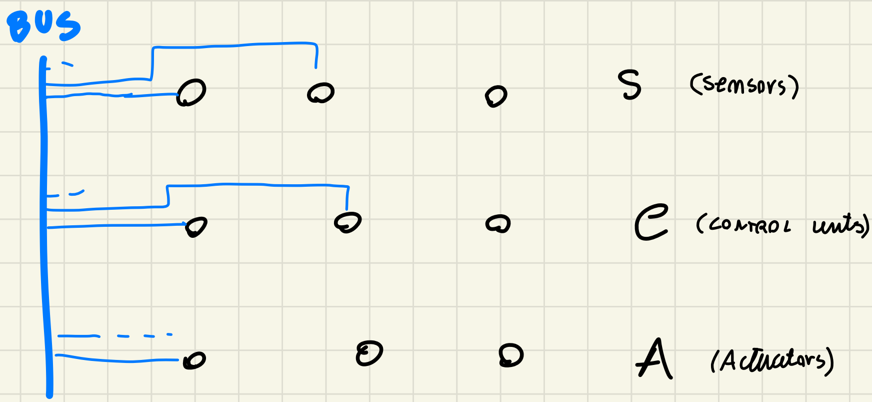
- **Control systems**:
  - **VCU (Vehicle Control Unit)**: High-level dynamic control system that manages active suspensions, ESP (electronic stability program) or ESC (electronic stability control).
    - Cruise control: maintains constant longitudinal speed
    - ESP: Helps improve cornering and control. It can sense differences between the driver's intentions and the vehicle's directions during turns.
    - TCS: Traction Control System. Prevents vehicle's wheels from spinning excessively while on slippery surfaces. Complementary to ABS.
  - **Dashboard display** control system: mainly for processing and displaying data (for the driver) from sensors
  - **Anti-collision assist**
  - **Lane warning**
  - **Air conditioning**.

↓ too many CONTROL system to increase modularity and decrease complexity each made by different company

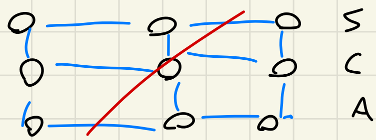


(CAR)

more elements of the system → LOT of small electronic devices



1) you can wire each device to let them communcate,
BUT too many issues, maintenance, cost..



wire all connections... NO!

2) simple BUS → shared BUS, one info at time, they have to wait for their turn someway

to let them communicate with each other, shared BUS, all connected to the BUS, one wire bringing one packet at time..

Examples

Control systems in cars

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Why so many different hardware control systems? To reduce complexity, increase modularity/maintanance and since they are generally produced from different companies!

Examples

Control systems in cars

- All these elements (sensors, actuators, control units) must **talk** to each other!

- Wired **point-to-point connections are impractical**
  - Costly
  - Space consuming
  - Messy
  - Not reconfigurable (addition/replacement of devices)

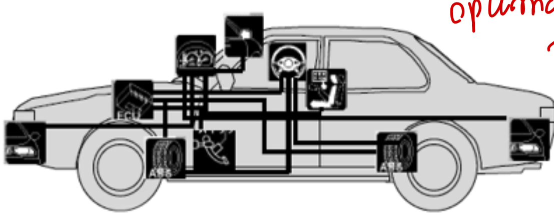
- Present solution: **serial bus** CAN (controller area network). Shared digital network.

common issues

- Communication problems, due to the shared network:
  - **Delays** (possibly non-bounded and non-constant)
  - **Package dropouts** ecc...

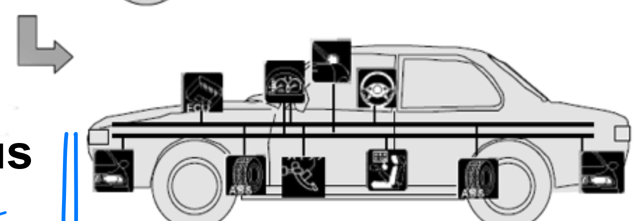
(Network issue for large modes communication)

optimal BUT unfeasible



Point-to-point connections

BAD



Serial bus

OK, all connect!

With new issues

Networked control

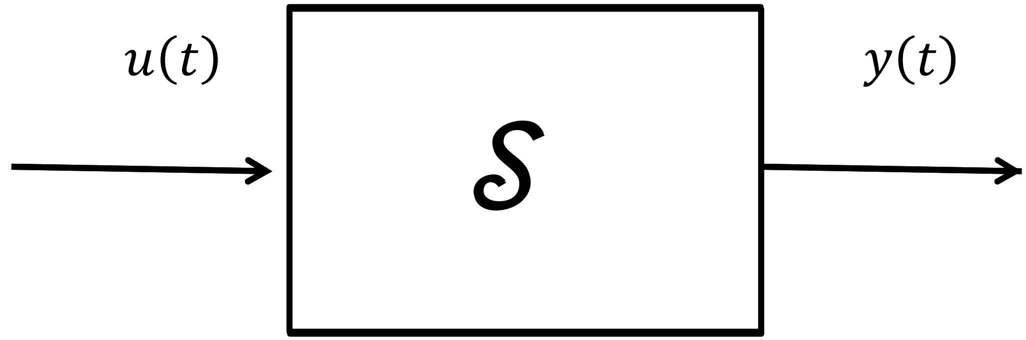
Sum-up: issues in control for network of components..

What are the common features of the examples?

(1) chemical plant, underwater, irrigation...
system of large # subsyst interconnected

MIMO systems

Very complex and large-scale system under control: large number of interacting subsystems



- Car
- Chemical plant
- Electric grid
- Fleet of autonomous underwater vehicles
- Irrigation network
- Manufacturing plant
- ...

- Need of modular and hierarchical control scheme
  - Subsystem/device level
  - Coordination between subsystems.

Issue: system of large # sub syst / large scaled to control,

respect centralized
↳ alternatives: DECENTRALIZED (subsystem level)
(allow communication) ← DISTRIBUTED, HYERARCHICAL → coordination

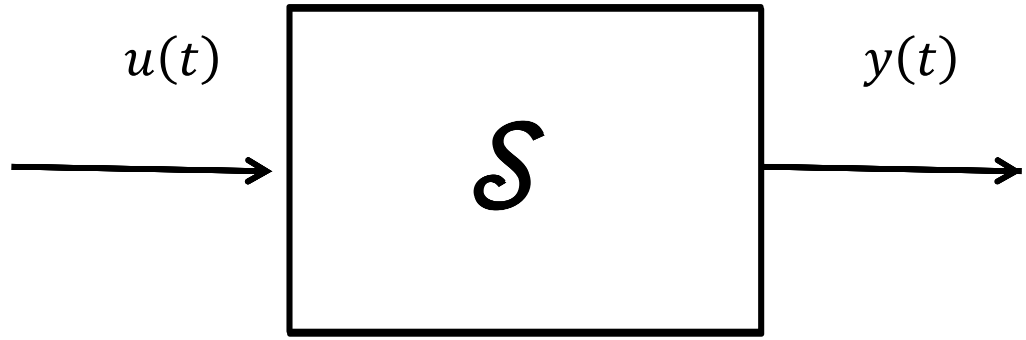
Networked control

What are the common features of the examples?

MIMO systems

Presence of a (possibly very) large number of sensors and actuators, possibly in different locations and geographically distributed

Presence of many units: communication issue



- Car
- Chemical plant
- Electric grid
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- ...

other issues system: wide # of parts that talk to each other..

lot of sensor/actuators.. that talk to each other → communication issue (connected to single bus)

Networked control

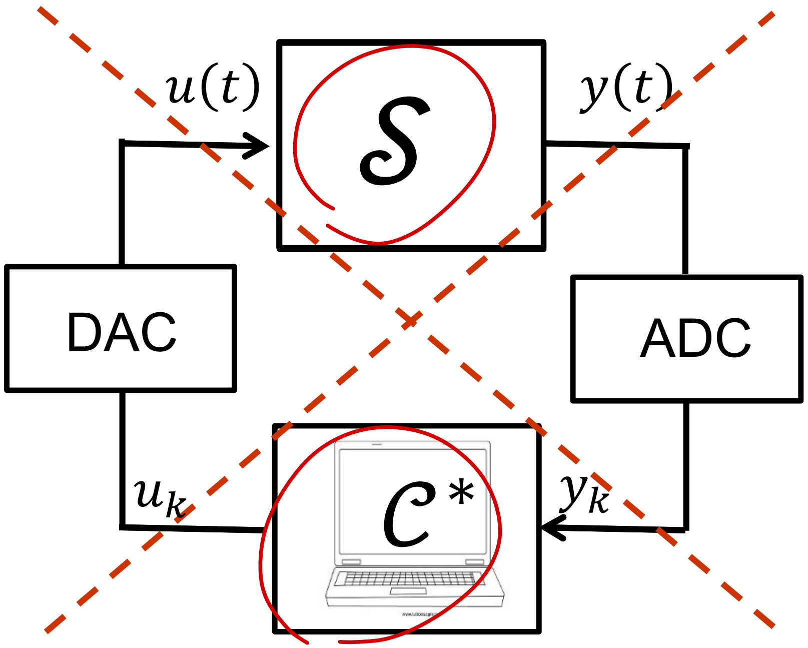
What are the common features of the examples?

New control applications pose new **challenges**.

↓ Problem solutions of CONTROL..
challenge!

Classic «centralized» control solutions with ideal communication may be not feasible/possible/adequate

Decentralized structure ecc.. solutions bringing challenges!



we know how to

Networked control

Technological advances:

Low-cost processing power.
Possibility of controlling at remote locations and/or at subsystem level.

Possibility of **transmitting information reliably via shared digital networks** (even wirelessly).

Emergence of new areas of research:

Complex control architectures (distributed, decentralized, hierarchical)

Control over dedicated (serial) communication networks

Networked control

This course will deal with these two topics:

- **Complex control architectures** (distributed and decentralized)
- **Control over** dedicated (serial) communication **networks**

NETWORKED ISSES

and in particular with their

How to design!
classical control is NOT enough

opportunities	challenges
- Flexibility - Reliability - Robustness - Reconfugurability - Scalability (computational, communication)	- Control challenges (design of distributed, decentralized, hierarchical controllers) - Network-induced challenges (bandwidth limitations, delays, jitter, packet dropouts) ← communications issues in a CONTROL NETWORK

Main related references

- G.P. Ferrari, E. Pizzi, Applicazione di schemi innovativi di controllo predittivo alla sezione di refrigerazione di un impianto per il trattamento del gas naturale, *MsC Thesis*, Politecnico di Milano (supervisor: Prof. M. Farina), 2013-2014.
- M. Banfi, Sviluppo e analisi di sistemi di controllo per l'inseguimento di segnali applicati a un impianto chimico di trattamento di gas naturale, *MsC Thesis*, Politecnico di Milano (supervisor: Prof. M. Farina), 2014-2015.
- R. J. Baillieul, P. J. Antsaklis. Control and communication challenges in networked real-time systems. *Proceedings of the IEEE*, 95 (1), pp. 9-28.
- J. R. Moyne and D. M. Tilbury. The emergence of industrial control networks for manufacturing control, diagnostics, and safety data. *Proceedings of the IEEE*, 95 (1), pp. 29-47.
- N. E. Leonard *et al*. Collective motion, sensor networks, and ocean sampling. *Proceedings of the IEEE*, 95 (1), pp. 48-74.
- M. Cantoni *et al*. Control of large-scale irrigation networks. *Proceedings of the IEEE*, 95 (1), pp. 75-91.

→ NEXT.. complex CONTROL Architecture, distributed /decentralized

how to design...

- decompose the model

than design controllers

- FIRST preliminaries!